ESP32 BT & WI-FI SECURITY SUITE

Complete Step-by-Step Project Setup & Build Guide

LEGAL & EDUCATIONAL DISCLAIMER:
This device design and documentation are strictly for educational and authorized penetration testing in controlled environments. Operating RF jammers or packet injection without authorization is illegal.

1. Bill of Materials (Required Hardware)

Component Name	Specification / Details	Quantity / Purpose
ESP32 Dev Board	ESP32-WROOM-32 / NodeMCU-32S (30-Pin)	1x Main Microcontroller
2.4" TFT LCD Shield	MCUFRIEND 8-bit Parallel (Red PCB)	1x Touchscreen Display
NRF24L01 Modules	2.4GHz Transceiver (Built-in PCB Antenna)	2x Dual RF Hopping Radios
MicroSD Card	FAT32 Formatted (up to 32GB)	1x Offline Wordlist Storage
Tactile Pushbuttons	6x6mm Pushbutton Switches	2x Physical Controls (ON/OFF)
Breadboard & Jumpers	830-Point Breadboard + F-M / M-M Wires	1x Prototype Wiring Assembly
5V USB Power Source	5V 1A - 2A Power Bank or USB Cable	1x Power Supply

2. Step 1: Hardware Wiring Overview

Connect all components according to the master pin allocation schema:

Interface / Device	ESP32 Pins Connected	Wiring Description
Radio 1 (NRF24 #1)	CE: GPIO 22 | CSN: GPIO 21	Lower Channel Range (2402 - 2440 MHz)
Radio 2 (NRF24 #2)	CE: GPIO 16 | CSN: GPIO 15	Upper Channel Range (2441 - 2480 MHz)
Shared SPI Bus	SCK: GPIO 18 | MOSI: GPIO 23 | MISO: GPIO 19	Connected to Radio 1, Radio 2, & SD Card
TFT LCD Bus	LCD_RST: 33, CS: 5, RS: 4, WR: 2	Data Bus D0-D7: GPIO 12, 13, 26, 25, 17, 27, 14, 32
Physical Buttons	TURN ON: GPIO 12 | TURN OFF: GPIO 14	One leg to GPIO pin, other leg to GND Rail
Power Connections	VCC: 3.3V Rail | Display Power: 5V/3.3V	Common Ground (GND) Rail connected to all devices

3. Step 2: MicroSD Card Preparation

1. Insert MicroSD card into your computer and format as FAT32.
2. Open the SD_Files/ folder in the project repository.
3. Copy wordlist.txt to the root directory of your MicroSD card (e.g. E:/wordlist.txt).
4. Insert the MicroSD card into the SD slot on the back of the 2.4" TFT Display Shield.

4. Step 3: Arduino IDE Configuration & Libraries

1. Download & Install Arduino IDE (v2.0 or newer recommended from arduino.cc).
2. Open Arduino IDE -> File -> Preferences.
3. In Additional Boards Manager URLs, paste the ESP32 package link:
   https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
4. Go to Tools -> Board -> Boards Manager..., search for "esp32" by Espressif Systems and click Install.
5. Open Library Manager (Tools -> Manage Libraries...) and install the following required libraries:

Library Name	Author	Purpose
MCUFRIEND_kbv	David Prentice	8-Bit Parallel TFT LCD Driver
TouchScreen	Adafruit	Resistive Touchscreen Reading
RF24	TMRh20	NRF24L01 2.4GHz Transceiver Control
Adafruit GFX Library	Adafruit	Core Graphics Drawing Primitives

5. Step 4: Compiling & Uploading Firmware

1. Open BluetoothJammer/BluetoothJammer.ino in Arduino IDE.
2. Select Board: Tools -> Board -> ESP32 Arduino -> ESP32 Dev Module.
3. Select COM Port: Tools -> Port -> Select your connected ESP32 serial COM port.
4. Click Upload (Right Arrow button).

Note: If uploading pauses at "Connecting...", press & hold the physical BOOT button (GPIO 0) on the ESP32.

6. Step 5: System Operation & UI Controls

On power-up, the device renders a 3D rotating wireframe cube boot animation and enters the Main Menu:

UI Menu Option	Function / Description	Controls
1. WIFI SCANNER	Sub-menu with 1.1 SELECT WIFI & 1.2 FIND PASSWORD options	Touch screen or TURN ON btn
2. WIFI+BT JAMMER	Dual NRF24 RF signal hopping (2.4GHz BLE & BT Classic)	Touch JAMMER ON / OFF
3. RESET	Restart device to initial state	Touch YES / NO confirmation

Additional Interface Controls:

- BOOT Button (GPIO 0): Acts as BACK key to navigate to previous menu.
- Web Portal (http://192.168.4.1): Connect Wi-Fi to "ESP32-Security-Tool" (Pass: 12345678) for web control.
- Serial CLI: Open Serial Monitor at 115200 baud (Commands: scan, deauth, hashcat).